



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization experienced an ICMP flood DoS attack that made internal network services unavailable for about two hours. The attack succeeded because the firewall was not properly configured to control incoming ICMP traffic. The security team contained the attack by blocking ICMP packets, taking non-critical services offline, and restoring critical services. Afterward, they strengthened security by adding ICMP rate limiting, source IP verification, network monitoring, and IDS/IPS filtering.
Identify	The organization should regularly audit the internal network, firewall configurations, systems, and access controls to identify vulnerabilities. In this incident, the main weakness was an improperly configured firewall that allowed a large volume of ICMP traffic to enter the network without restriction.
Protect	The organization should strengthen firewall configurations by limiting the rate of incoming ICMP packets and verifying source IP addresses to reduce spoofed traffic. Security policies, system hardening, and regular configuration reviews should also be used to reduce exposure to similar attacks.
Detect	The organization should use network monitoring tools and IDS/IPS systems to detect abnormal traffic patterns, such as sudden spikes in ICMP packets. Alerts should be configured so the security team can respond quickly when unusual

	activity is detected.
Respond	During the incident, the security team should contain the attack by blocking malicious ICMP traffic, taking non-critical services offline if necessary, and prioritizing critical network services. The incident should then be investigated to determine the cause, document findings, and improve response procedures.
Recover	The organization should restore affected network services to normal operation, verify that systems are functioning properly, and bring non-critical services back online. After recovery, the team should review the incident, update firewall rules and security procedures, and use the lessons learned to improve future resilience.

Reflections/Notes:

This activity helped me understand that incident response is not only about stopping an attack while it is happening. A security team also needs to identify why the attack was successful, improve preventive controls, monitor for similar behavior, and restore operations safely.

The NIST CSF provides a clear structure for handling this process. In this scenario, the unconfigured firewall was the main weakness, while controls such as rate limiting, source IP verification, and IDS/IPS monitoring helped reduce the chance and impact of another ICMP flood.